Build the edge.
Then prove it can execute.

A small quant team building low-latency crypto arbitrage systems, from market data and research to simulation and execution.

01 / THE OPPORTUNITY

Crypto liquidity is split across venues, chains and instruments. We will **test whether observable price discrepancies survive actual execution costs**, rather than treating a quoted spread as a trading profit. The goal is a working research-to-execution pipeline, not another notebook that assumes every order fills.

02 / RESEARCH SCOPE

On-chain arbitrage and MEV. Same-chain price gaps, simulated backruns and transaction-ordering research. Front-running is a research topic, not an approved live strategy: no confidential/client order flow, manipulation or sandwiching.

Spot and perpetuals. Study cross-venue spreads, spot/perp basis and funding-aware hedges. Account for margin, hedge failures and liquidation risk.

Prediction markets. Test inconsistencies within complete outcome sets and between genuinely equivalent contracts. Matching titles are not enough: resolution rules, deadlines and settlement must match.

Cross-chain arbitrage. Study pre-positioned inventory and later rebalancing. Separate-chain fills are not one atomic trade; bridge delays, capital usage and temporary exposure belong in the model.

03 / START NARROW, EXPAND ON EVIDENCE

First milestone: select one liquid pair and one route family, capture synchronised data, and build a replayable executable-edge benchmark. Choose between a same-chain route and a tightly scoped spot/perp route after a short data/access review. Prediction markets and cross-chain routes are extensions, not four simultaneous builds.

What we want to demonstrate
A reproducible opportunity, realistic fill assumptions, measured latency and positive net edge after costs. HFT here means latency-sensitive crypto execution, not a claim of institutional CEX-colocation performance.

5-6 people. Clear ownership.

The team is being formed. Ryad coordinates the project; the other roles are open. We are looking for complementary builders, not five people doing the same research in parallel.

04 / WHO WE NEED

Project / integration. Ryad: scope, architecture, weekly review and integration.

2 quant researchers. Microstructure / perps and prediction-market / relative-value research.

1 execution engineer. Rust, C++ or Solidity; adapters, simulation, routing and transaction lifecycle.

1 data / research engineer. Python or SQL; timestamped data, event replay, reproducibility and diagnostics.

Optional 6th member. Low-latency systems, infrastructure, monitoring or risk tooling.

05 / PROPOSED SIX-WEEK FIRST CYCLE

Week 1 - Scope and data. Agree one hypothesis, instruments, data quality checks, access constraints and a shared interface.

Weeks 2-3 - Replay and falsification. Measure executable prices, fees, funding/borrow where relevant, failed or partial fills and hedge/rebalancing costs. Test on held-out periods, not only favourable examples.

Weeks 4-5 - Live observation. Run shadow execution without risking funds; log opportunity decay, submission/inclusion latency, fill assumptions and exposure.

Week 6 - Review and decision. Produce a reproducible report and demo. A small live pilot is conditional on results, venue/legal eligibility, security checks and explicitly agreed capital and loss limits.

06 / HOW WE WORK

One shared pipeline: **data -> signal -> cost/risk filter -> simulation -> execution -> attribution**. Python for research; compiled execution only where benchmarks justify it. Each workstream has an owner, a test and a weekly deliverable. Suggestion: one short weekly review and a focused build session; agree availability, ownership and any economic terms at kickoff. No private employer code or credentials enter the project.

Interested? DM Ryad in Fund with your preferred role, one relevant project or skill, and realistic weekly availability. Strong coding or research fundamentals matter more than already being a crypto expert.

TECHNICAL REFERENCE POINTS

Flashbots: backrunning and bundles
Hyperliquid: funding mechanics
Polymarket: resolution rules